

Venkata Saratchandra Patnaik Markonda Patnaikuni

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EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ

TECHNICAL SKILLS

Programming & Scripting: Python, C, Java, R, SQL, Shell scripting

Industrial Automation & Embedded Systems: IoT, NodeMCU ESP8266, PLC simulation, HMI/SCADA concepts, real-time monitoring, motion control principles

Cloud & DevOps Tools: AWS (S3, EC2, Lambda, Greengrass, EKS), Docker, Kubernetes, CI/CD pipelines (GitOps, ArgoCD, Jenkins)

Databases: MySQL, PostgreSQL, MariaDB, Firebase

Analytical & Research Tools: Pandas, NumPy, Matplotlib, PyTorch, TensorFlow, OpenAI API

Project Management & Collaboration: Jira, GitHub, Microsoft Project

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- Integrated StreamDeck client systems for real-time control of media dashboards, enabling operational efficiency in live broadcast workflows.
- Designed automation workflows using Terraform and Python scripts to streamline onboarding, reducing manual setup time by 90%.
- Managed 15+ containerized microservices on AWS EKS with Docker/Kubernetes; deployed CI/CD pipelines via GitOps and ArgoCD, improving reliability and reducing failures by 95%.
- Developed monitoring and diagnostic scripts for real-time systems, simulating industrial control and automation processes.

Web Development Intern: Blueplanet Solutions Inc., India

April 2021 - June 2021

- Optimized MySQL database using stored procedures for 'Campus Club', improving query speed by 50% and enabling faster data access for over 1,000+ student profiles.
- Enhanced a responsive student search UI with HTML, CSS, and JavaScript, reducing search time by 60% and improving user experience.
- Analyzed web server logs to identify memory leaks, enhancing application stability by 30% and reducing server resource usage.

PROJECTS

Underground Cable Fault Detection (IoT and Embedded Systems)

- Designed and simulated a fault detection system for underground cables using NodeMCU ESP8266 and voltage sensing, implementing real-time monitoring with LCD and remote notifications.
- Applied embedded programming and control logic, simulating industrial automation workflows and system diagnostics.
- Integrated automation principles, including fault detection, alerting, and real-time response, demonstrating transferable PLC/HMI/SCADA skills.

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Built an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged a complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda. Automated deployments via shell scripts and passed over 100+ real-time autograder test cases.